CS4048 Data Science

Project Deliverable 1

<u>Roll No 1:</u> 22L-6562

<u>Name 1:</u> Husnain Ali

<u>Section:</u> 7A

<u>Roll No 2:</u> 22L-6552

<u>Name 2:</u> Muhammad Faizan Shabir

<u>Section:</u> 7A

Dataset: Description and source

- (Columbia Gaze Data Set) Available at : https://www.cs.columbia.edu/CAVE/databases/columbia_gaze/

- Data set contains a total of 5,880 high-resolution images of 56 different people (32 male, 24 female).
- Each image has a resolution of 5,184 x 3,456 pixels.
- 21 of the subjects were Asian, 19 were White, 8 were South Asian, 7 were Black, and 4 were Hispanic or Latino.
- The subjects ranged from 18 to 36 years of age, and 21 of them wore prescription glasses.
- For each subject, images were acquired for each combination of five horizontal head poses (0°, ±15°, ±30°), seven horizontal gaze directions (0°, ±5°, ±10°, ±15°), and three vertical gaze directions (0°, ±10°).
- Total of 5 * 7 * 3 = 105 combinations for each subject.

Preprocessing Techniques: Methods used and their intuition

- Resized the images to 224 by 224 since the MobileNet CNN model requires images to be of this size.
- Pixel values for images were normalized to [0, 1] since MobileNet CNN model requires pixel values to be in this range.
- The head pose, horizontal gaze and vertical gaze angles were extracted from image file names and images were labelled as ‘attentive’, ‘distracted’, or ‘disengaged’ based on angle thresholds set by us. This was done to label the dataset for supervised learning.

Visualizations: Basic insights or distributions

- 5 sample images were outputted for each subject to visualize how images were labeled as ‘attentive’, ‘distracted’, or ‘disengaged’.
- A bar chart was made to see the number of images in each class (‘attentive’, ‘distracted’, or ‘disengaged’). It was observed that there is a class imbalance with 2688 images labelled as ‘attentive’, 2520 images labelled as ‘distracted’ and 672 images labelled as ‘disengaged’.

Future Work: Planned next steps for model implementation

- It is planned to fine-tune MobileNet CNN on the dataset.